Machine Learning

Dr. Shylaja S S

Professor

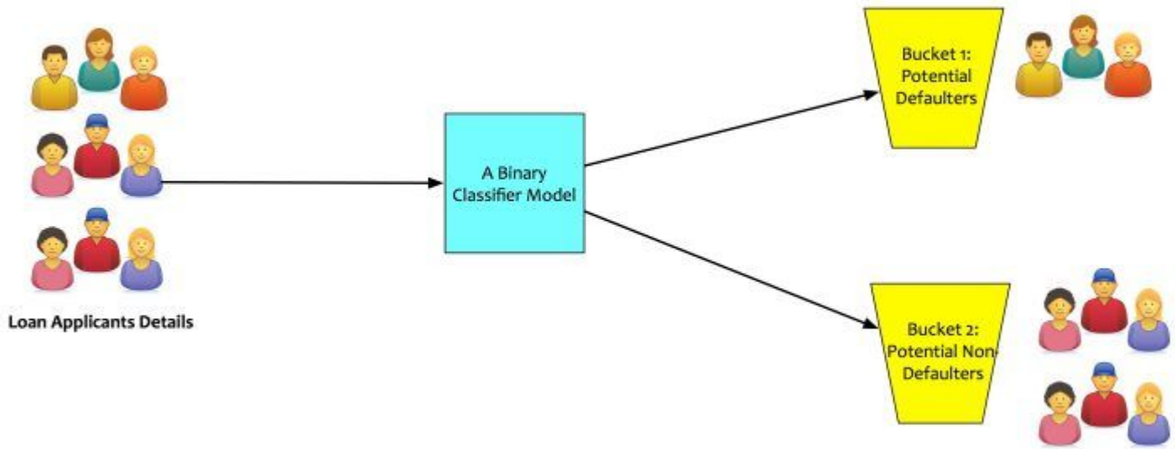
Department of Computer Science & Engineering

Acknowledgement : Andrew Ng (Stanford University)
Dr. Rajanikanth K (Former Principal MSRIT, PhD(IISc),
Academic Advisor, PESU)
Teaching Assistant (Arya Rajiv Chaloli)

Classification Problem

Binary Classification:

Email: Spam / Not-Spam
Tumor: Malignant / Benign



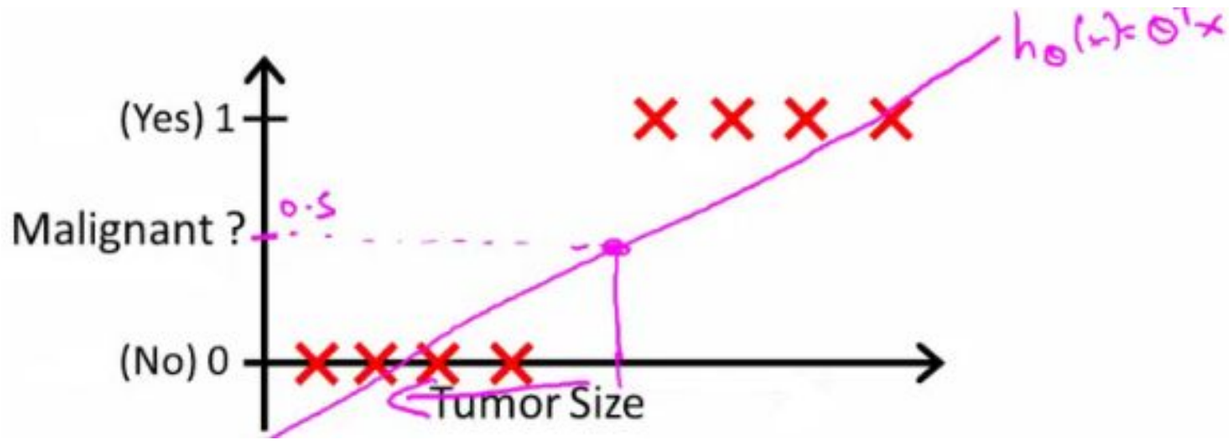
$y \in \{ 0,1 \}$
0: "negative class"
1: "positive class"

Multi-class classification problem: $y \in \{ 0,1,2,3\ldots \}$

Developing a classification algorithm

Problem Definition:
Tumour size vs Malignancy (0 or 1)

We could use linear regression



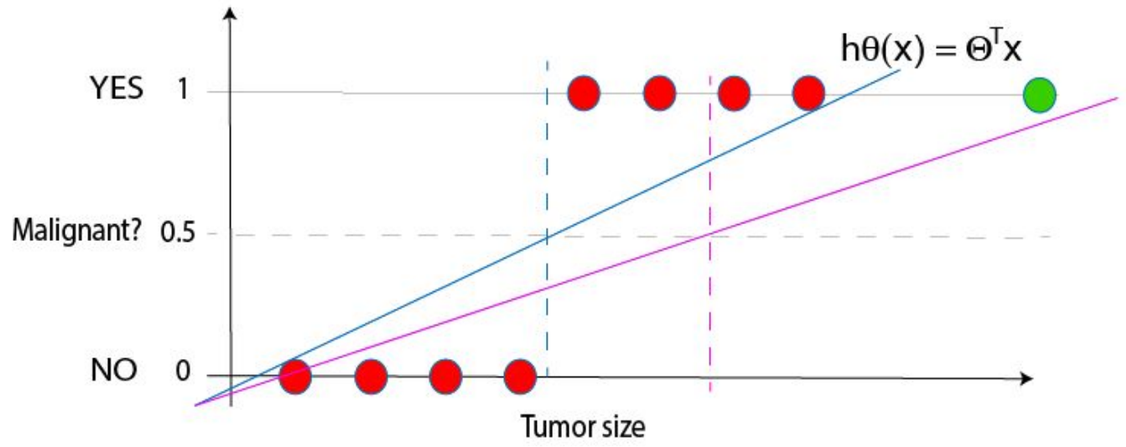
To make predictions the threshold classifier output:

- $h_\theta(x) >= 0.5$, predict y = 1
- $h_\theta(x) < 0.5$, predict y = 0

Developing a classification algorithm

Adding an extra point further away.

The linear regression model provides a bad fit.



- Often linear regression is not suitable for classification problems

- It gives a hypothesis function which can output a value much more than 1 or less than 0, when the true value is always either 0 or 1
  - $h_\theta(x) > 1$, $h_\theta(x) < 0$ are possible
  - but $y \in \{ 0,1 \}$

Thus, develop an algorithm which has output/predictions that are in the range of 0-1, i.e., $0 <= h_\theta(x) <= 1$

⇨ Logistic Regression

Hypothesis representation

The hypothesis (output) of a linear regression is

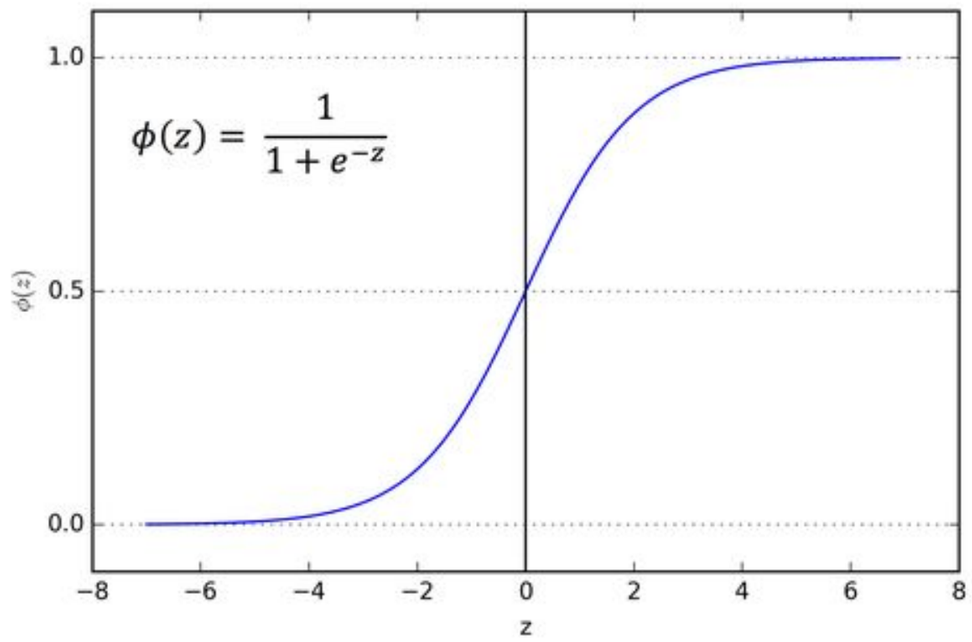
$$h_\theta(x) = \theta^T x$$

In logistic regression, for the hypothesis, we use the **sigmoid/logistic function** over the linear regression output to contain it within 0 and 1

$$h_\theta(x) = g(\theta^T x)$$

$$g(x) = \frac{1}{1+e^{-z}}$$

$$h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$$



Interpretation of the hypothesis output

- Hypothesis of logistic regression ($h_\theta(x)$) is the **estimated probability that y=1 on input x**

- From the previous example:
  $h_\theta(x) = 0.7 \rightarrow$ means there is a 70% chance that the tumor is malignant

- $h_\theta(x) = p(\ y=1 \mid x;\theta\ )$ implies "*given x and θ, $h_\theta(x)$ is the probability that y=1*"

- Since *y* can only be 0 or 1, $p(\ y=1 \mid x;\theta\ ) + p(\ y=0 \mid x;\theta\ ) = 1$

Interpretation of the hypothesis output

$h_\theta(x) \geq 0.5,\ predict\ y=1$

$g(z) \geq 0.5$

$(when\ z \geq 0)$

$h_\theta(x) = g(\theta^T x) \geq 0.5$

$(when\ \theta^T x \geq 0)$

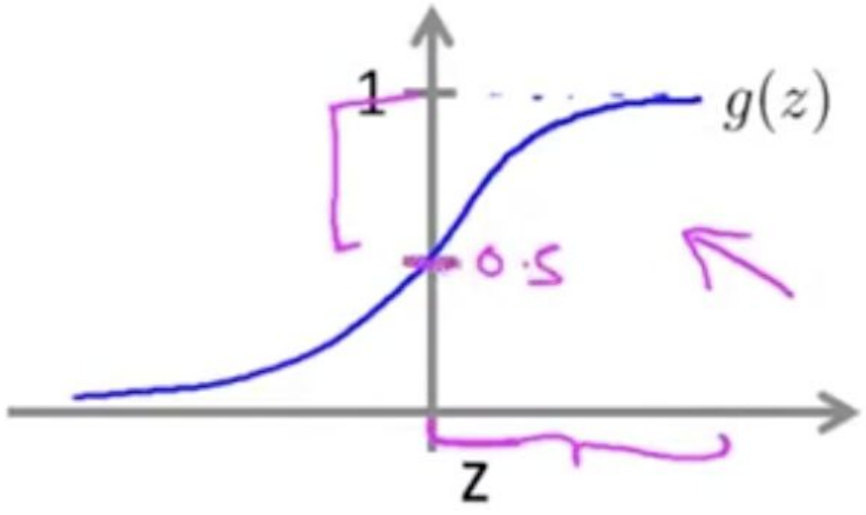
$h_\theta(x) < 0.5,\ predict\ y=0$

$g(z) < 0.5$

$(when\ z < 0)$

$h_\theta(x) = g(\theta^T x) < 0.5$

$(when\ \theta^T x < 0)$



Decision Boundary

- Let $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$
- Assume

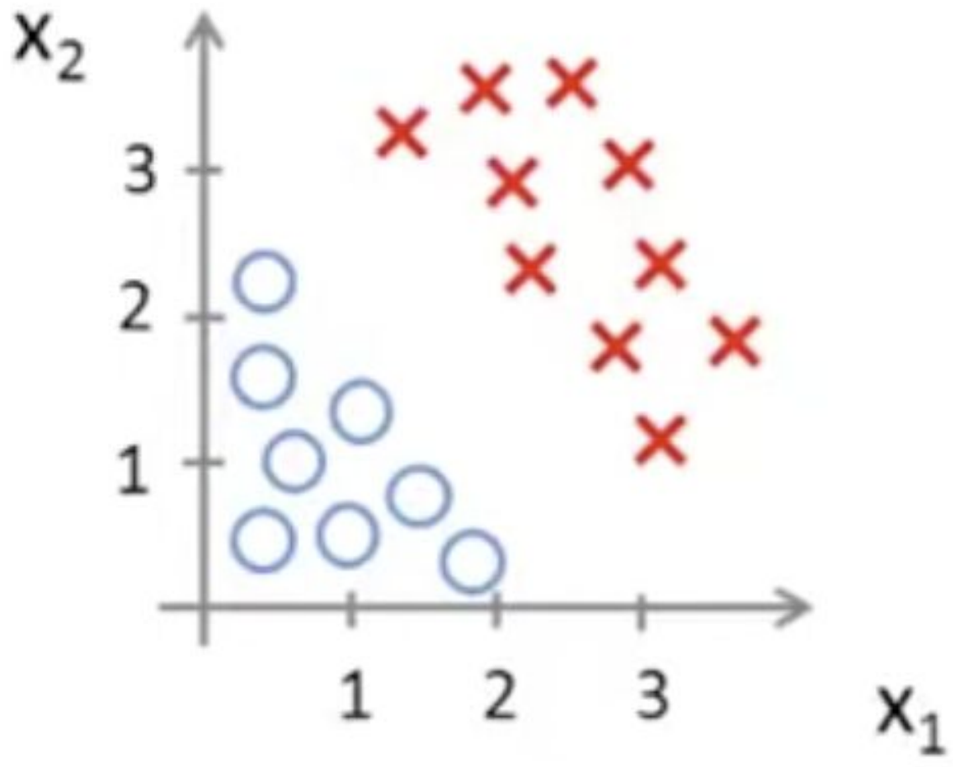
$$\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

- Predict y=1 if

…

- Predict y=0 if

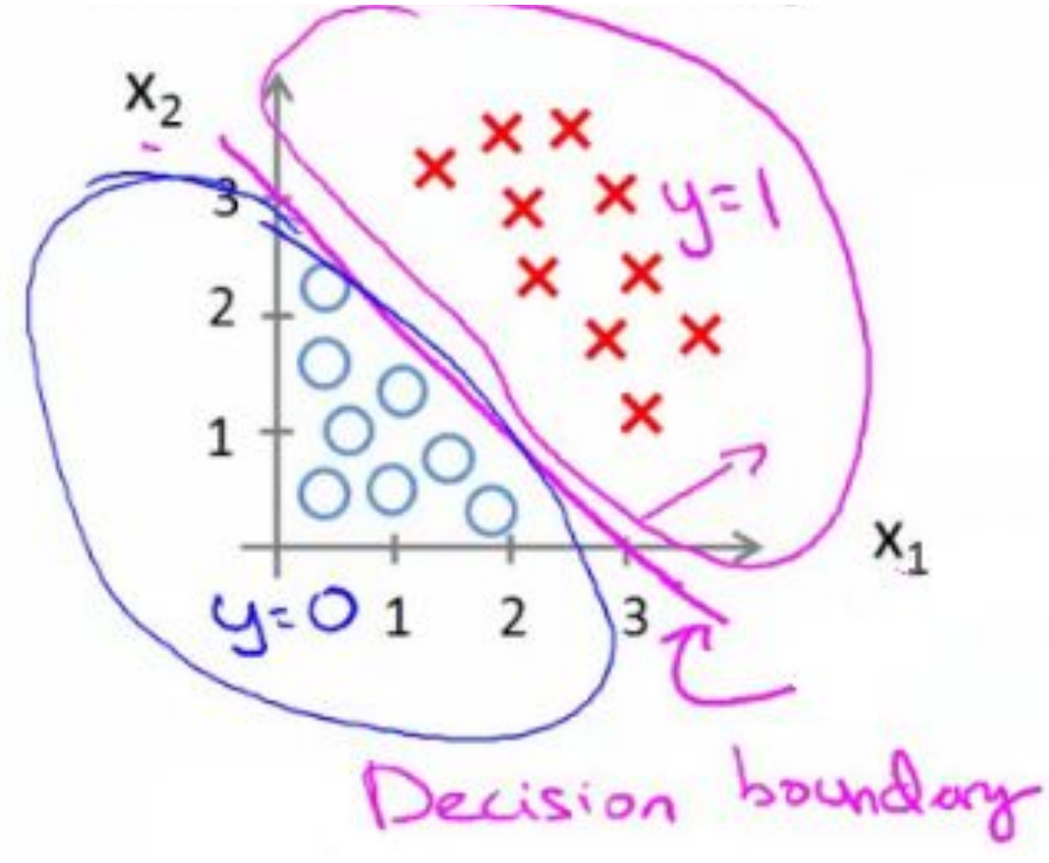
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Machine Learning

Decision Boundary

- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$
- $$\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$



- Predict y=1 if
  $-3 + x_1 + x_2 >= 0$
  $x_1 + x_2 >= 3$

- Similarly, predict y=0 if
  $-3 + x_1 + x_2 < 0$
  $x_1 + x_2 < 3$

x1 + x2 = 3 → graphically plot the **decision boundary**

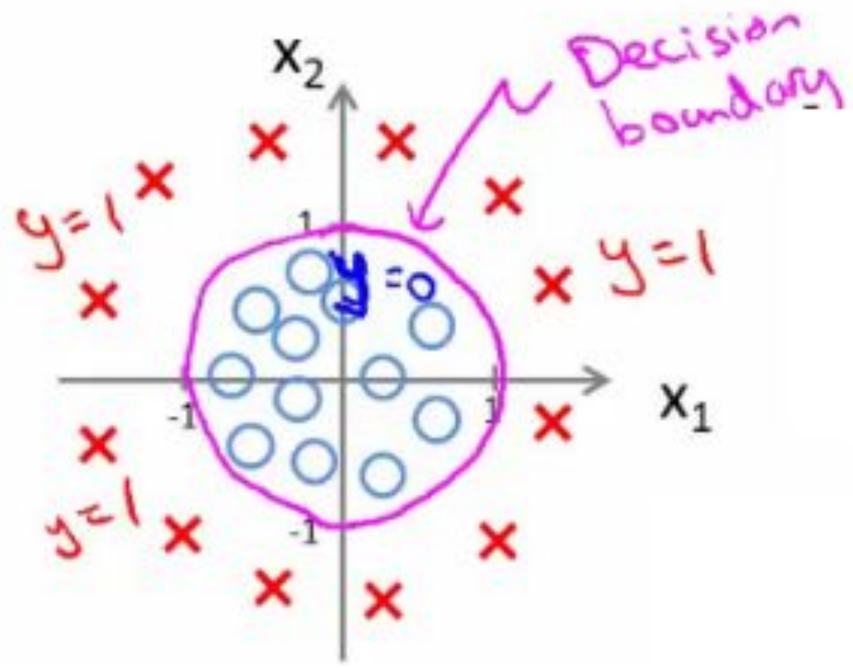
Non Linear Decision Boundary

- Get logistic regression to fit a complex non-linear data set
- Add higher order terms
- Let $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_3 x_1^2 + \theta_4 x_2^2)$
- Assume θ^T was [-1,0,0,1,1]
- Then predict y=1 if

  $-1 + x_1^2 + x_2^2 >= 0$

  or

  $x_1^2 + x_2^2 >= 1$



- Plot $x_1^2 + x_2^2 = 1$ gives us a circle with a radius of 1 around 0
- Higher order polynomial terms, yields even more complex decision boundaries

Cost function for fitting a logistic regression model

Defining the optimization object for the cost function used to fit the parameters

Given:

- Training set: m examples { $(x^{(1)},y^{(1)}), (x^{(2)},y^{(2)}), \ldots (x^{(m)},y^{(m)})$ }
- n-dimensions: feature vector contains (n+1) elements $\theta^T = [\ \theta_0\ \theta_1 \ldots \theta_n\ ]$
- $x_0 = 1$, $y \in \{ 0,1 \}$
- Hypothesis is based on parameters (θ)

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose the parameters (θ) ?

Cost function for fitting a logistic regression model

Linear regression uses the following function to determine θ

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{2} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

Writing the squared error term as *Cost()*

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \text{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

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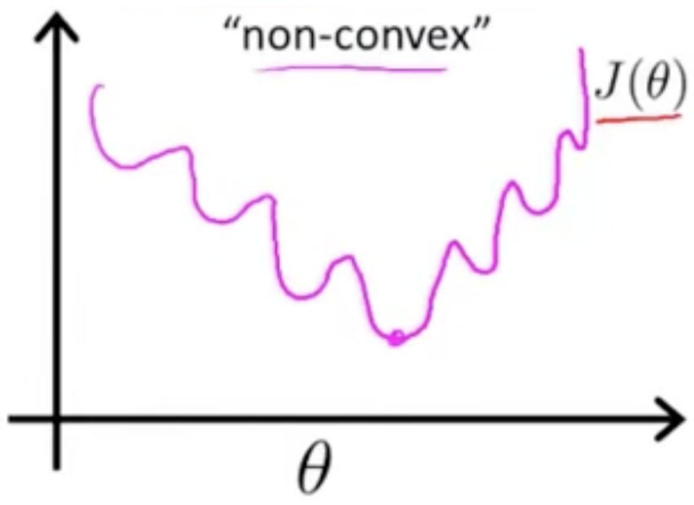
$$Cost\left(h_\theta(x), y \right) = \frac{1}{2} \left(h_\theta(x) - y \right)^2$$

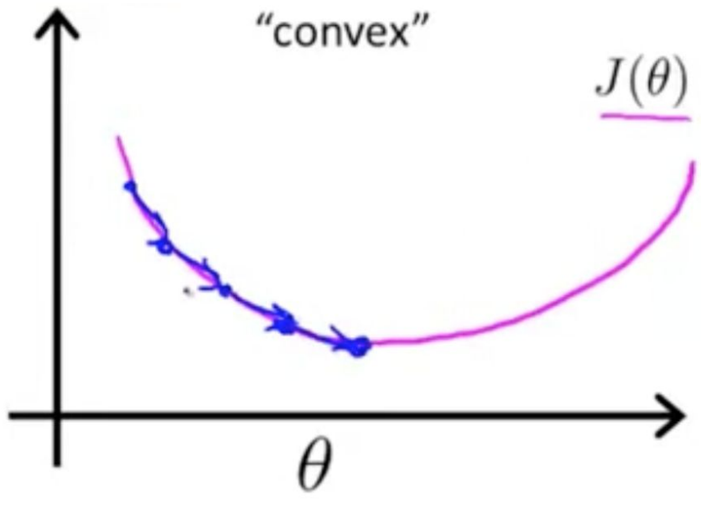
Cost function for fitting a logistic regression model

$$Cost\left(h_\theta(x), y \right) = \frac{1}{2}\left(h_\theta(x) - y \right)^2$$

In logistic regression, if we use this cost function, it will be a non-convex function of the parameters θ

This is because $h_\theta(x)$ is no longer a linear function. It is a complex non-linear function (sigmoid function).



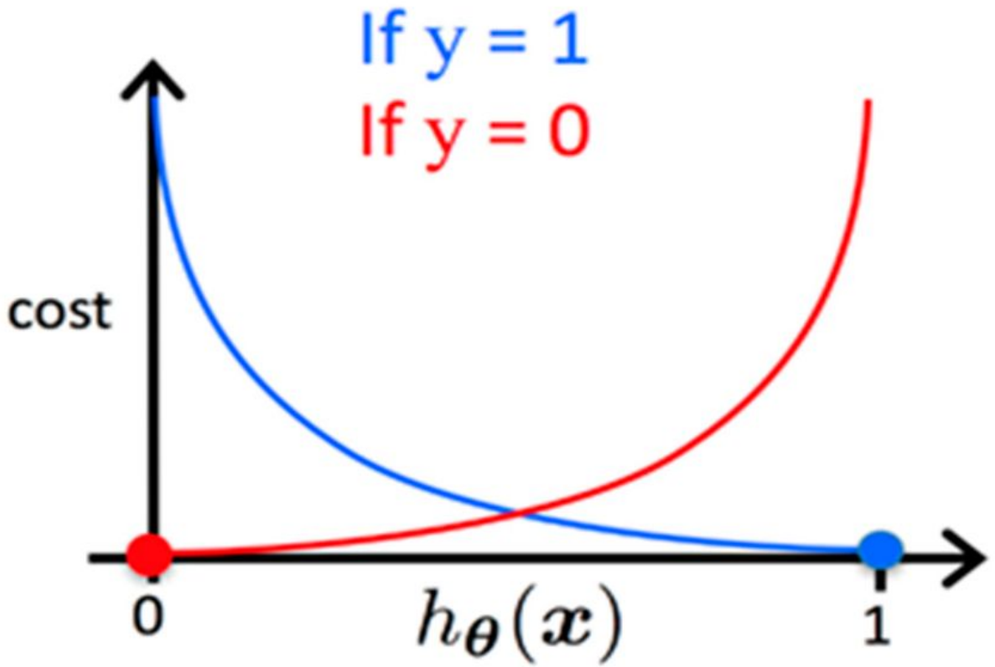


Lots of local minima means gradient descent may not find the global optimum

Cost function for fitting a logistic regression model

Thus a different cost function is required for logistic regression

$$\text{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$



This captures the intuition that if $h_\theta(x) = 0$ (predict $P(y=1|x; \theta) = 0$) but $y = 1$ this will penalize the learning algorithm with a massive cost

Following the previous example:
If the patient is told that the tumor is not malignant with full certainty ($h_\theta(x) = 0$), but the tumor was malignant in reality ($y=1$), then the cost, or the penalty is very high (∞)

Simplified Cost Function

The cost function needs to be simplified to apply gradient descent to find the optimal parameters

$$\text{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

It can be expressed as:

$$\text{Cost}(h_\theta(x),y) = - y \log(h_\theta(x)) - (1 - y) \log(1 - h_\theta(x))$$

Since y takes only 2 values {0,1} substituting the them shows that the simplified cost function is equivalent to the previous one

Simplified Cost Function

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \text{Cost}(h_\theta(x), y)$$

$$J(\theta) = -\frac{1}{m}[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_\theta(x^{(i)}))]$$

To fit/optimize parameters (θ): Minimize $J(\theta)$ wrt θ

To make a prediction given a new x: Output $h_\theta(x)=0$ with the optimized parameters θ

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Minimizing Cost Function using Gradient Descent

To minimize J(θ)

- Use gradient descent
- Repeatedly update each parameter using a learning rate

$$\text{Repeat } \{ \theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) \}$$

Repeat $\{$

$$\theta_j := \theta_j - \alpha \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

(simultaneously update all θ_j)

$\}$

Minimizing Cost Function using Gradient Descent

- This equation is the same as the linear regression rule
  - The only difference is that the definition for the hypothesis has changed

- When implementing logistic regression with gradient descent, we have to update all the θ values (θ_0 to θ_n) simultaneously
  - Could use a for loop
  - Better would be a vectorized implementation

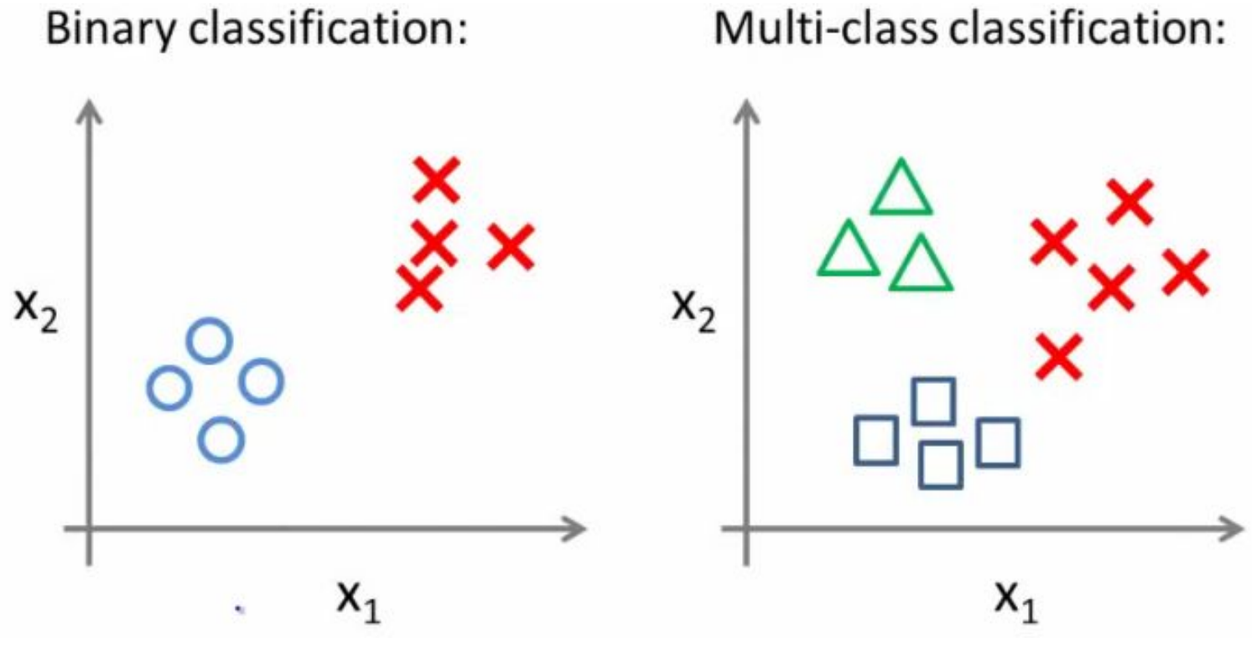
- Feature scaling for logistic regression helps run the gradient descent faster

Multi-Class Classification

Multi-class classification

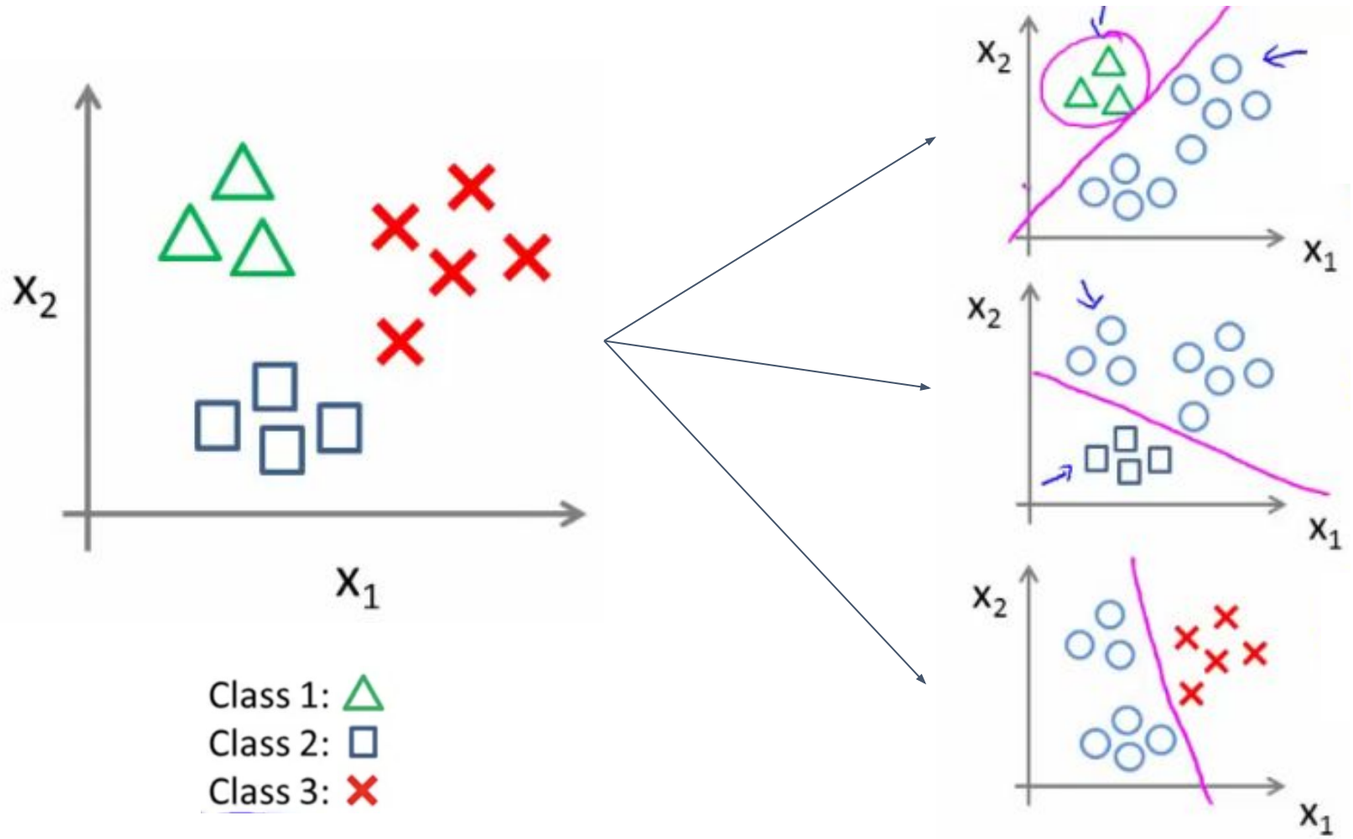
Email foldering and tagging: work / friends / family / hobby
Medical report: not- ill / flu / cold
Weather: sunny / cloudy / rain / snow

$y \in \{1,2,3\ldots\}$
1: "class 1"
2: "class 2"
3: "class 3"
… so on.



Multi class classification: One-vs-All

- Train a logistic regression classifier $h_\theta^{(i)}(x)$ for each class i to predict the probability that y = i
- On a new input, x to make a prediction, pick the class i that maximizes the probability that $h_\theta^{(i)}(x) = 1$



- $h_\theta^1(x) = 1$: $P(y=1 \mid x_1; \theta)$
- $h_\theta^2(x) = 1$: $P(y=1 \mid x_2; \theta)$
- $h_\theta^3(x) = 1$: $P(y=1 \mid x_3; \theta)$

Quiz

Suppose that you have trained a logistic regression classifier, and it outputs on a new example x a prediction $h_\theta(x) = 0.2$. This means (check all that apply):

a. Our estimate for $P(y = 1|x;\theta)$ is 0.8
b. Our estimate for $P(y = 0|x;\theta)$ is 0.8
c. Our estimate for $P(y = 1|x;\theta)$ is 0.2
d. Our estimate for $P(y = 0|x;\theta)$ is 0.2

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c. Our estimate for $P(y = 1|x;\theta)$ is 0.2
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Quiz

Which of the following are true (check all that apply):

a. J(θ) will be a convex function, so gradient descent should converge to a global minimum

b. Adding polynomial features (e.g., instead using $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_2 + \theta_4 x_1 x_2 + \theta_5 x_2)$) could increase how well we can fit the training data

c. The positive and negative examples cannot be separated using a straight line. So, gradient descent will fail to converge.

d. Because the positive and negative examples cannot be separated using a straight line, linear regression will perform as well as logistic regression on this data.

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For logistic regression, the gradient is given by $\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$. Which of these is a correct gradient descent update for logistic regression with a learning rate of α? Check all that apply.

a. $\theta := \theta - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(\theta^T x - y^{(i)}\right) x^{(i)}$.

b. $\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(\theta^T x - y^{(i)}\right) x_j^{(i)}$ (simultaneously update for all j).

c. $\theta := \theta - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x^{(i)}$.

d. $\theta := \theta - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(\frac{1}{1+e^{-\theta^T x^{(i)}}} - y^{(i)}\right) x^{(i)}$.

Machine Learning

Quiz

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- [x] $\theta := \theta - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x^{(i)}$.
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Quiz

Which of the following statements are true (check all that apply):

a. The cost function J(θ) for logistic regression trained with m≥1 examples is always greater than or equal to zero.
b. Linear regression always works well for classification if you classify by using a threshold on the prediction made by linear regression.
c. The sigmoid function is never greater than one
d. For logistic regression, sometimes gradient descent will converge to a local minimum (and fail to find a global minimum). (This is the reason we prefer more advanced optimisation algorithms like conjugate gradient/BFGS/L-BFGS etc)

Quiz

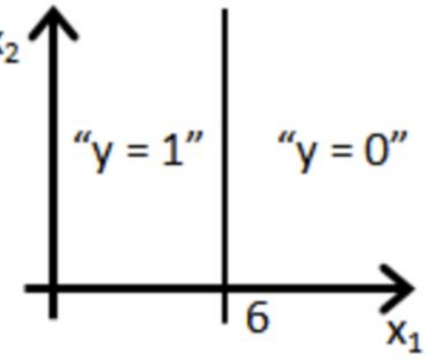
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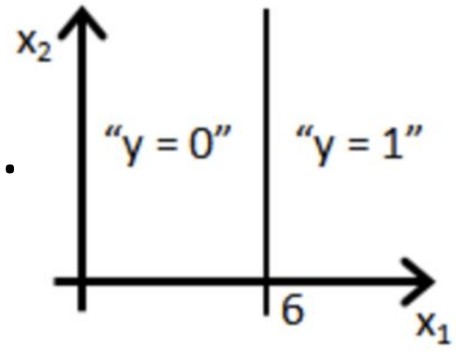
Quiz

Suppose you train a logistic classifier $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$. Suppose $\theta_0 = 6, \theta_1 = 0, \theta_2 = -1$. Which of the following figures represents the decision boundary found by your classifier?

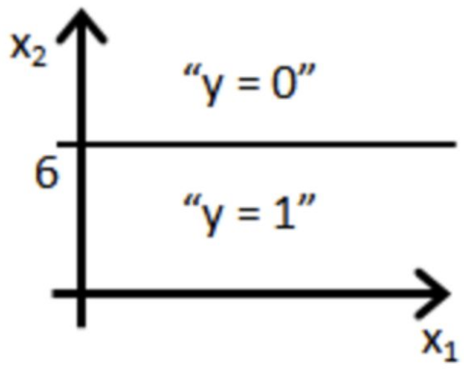
a.



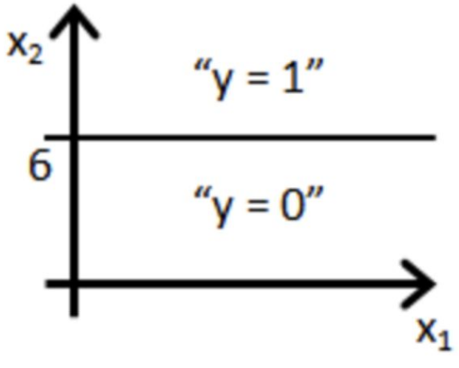
b.



c.



d.

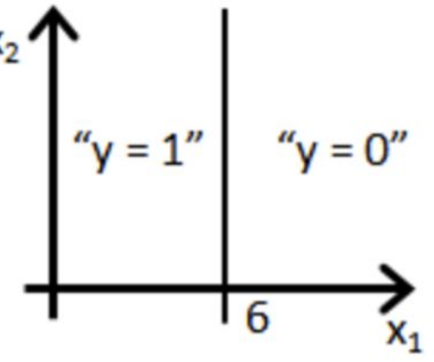


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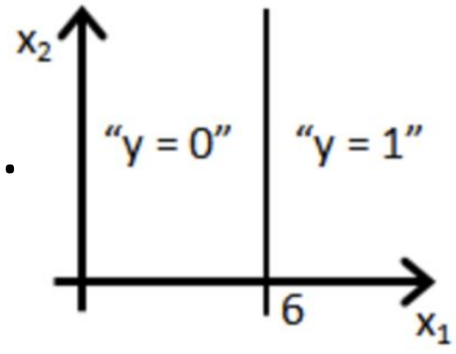
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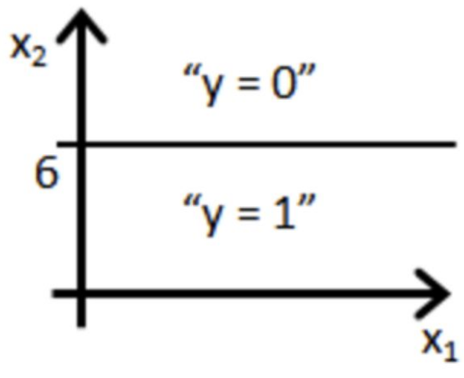
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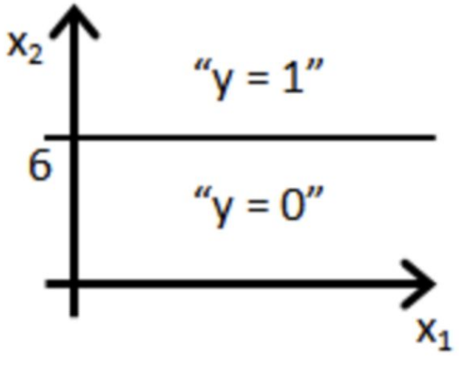
b.



c.



d.



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QUIZ Links

https://www.sanfoundry.com/artificial-intelligence-ai-questions-answers-online-quiz/#:~:text=Explanation%3A%20In%20automatic%20vehicle%20set,an%20example%20of%20supervised%20learning.

(https://thecodingshef.com/all-unit-mcq-questions-of-machine-learning/)

https://github.com/mGalarnyk/datasciencecoursera/blob/master/Stanford_Machine_Learning/Week3/week3quiz1.md

Machine Learning

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